Proof of Scaling Parameter Invariance

Assumptions:

1. C is a valid copula on $[0,1]^2$. By definition, $C(u_1, u_2)$ is non-decreasing in each of its coordinates.
2. The limit defining the generalized lower-tail dependence coefficient exists for at least one strictly positive scaling parameter.

Proposition 1. *For a copula C on $[0,1]^2$ and parameters $\gamma > 0$, $\delta \in (0,2)$, define*

$$\kappa_L(\delta, \gamma; C) = \lim_{u \downarrow 0} \frac{\log C\big(\gamma u^\delta, \gamma^{-1} u^{2-\delta}\big)}{\log u},$$

whenever the limit exists. Under the stated assumptions, $\kappa_L(\delta, \gamma; C)$ does not depend on the choice of the scaling parameter γ for all $\delta \in (0,2)$.

Proof. Let $\delta \in (0,2)$ and choose two arbitrary scaling parameters $\gamma_1, \gamma_2 > 0$. Suppose the limit is well-defined for γ_2, and denote it by λ:

$$\lambda = \kappa_L(\delta, \gamma_2; C) = \lim_{u \downarrow 0} \frac{\log C\big(\gamma_2 u^\delta, \gamma_2^{-1} u^{2-\delta}\big)}{\log u}.$$

We aim to prove that $\kappa_L(\delta, \gamma_1; C) = \lambda$, demonstrating that the measure is independent of the choice of the scaling parameter.

Step 1: Establishing Coordinate-wise Bounds

Define the ratio of the scaling parameters as $\rho = \frac{\gamma_1}{\gamma_2} > 0$. We then define two bounding constants based on this ratio:

$$a = \min\Big\{\rho^{1/\delta},\, \rho^{-1/(2-\delta)}\Big\}, \qquad b = \max\Big\{\rho^{1/\delta},\, \rho^{-1/(2-\delta)}\Big\}.$$

Because $\rho > 0$, both a and b are strictly positive.

For any $u > 0$, we can compare the input arguments for the copula function. By the definition of a, we know $a \le \rho^{1/\delta}$. Raising both sides to the power of δ, we get $a^\delta \le \rho = \frac{\gamma_1}{\gamma_2}$, which implies:

$$\gamma_2 (au)^\delta = \gamma_2 a^\delta u^\delta \le \gamma_1 u^\delta.$$

Similarly, since $a \le \rho^{-1/(2-\delta)}$, raising both sides to the power of $2-\delta$ yields $a^{2-\delta} \le \rho^{-1} = \frac{\gamma_2}{\gamma_1}$, giving:

$$\gamma_2^{-1} (au)^{2-\delta} = \gamma_2^{-1} a^{2-\delta} u^{2-\delta} \le \gamma_1^{-1} u^{2-\delta}.$$

By symmetric logic for the maximum boundary b, we have $b \ge \rho^{1/\delta}$ and $b \ge \rho^{-1/(2-\delta)}$, which produces the upper bounds:

$$\gamma_2 (bu)^\delta \ge \gamma_1 u^\delta, \qquad \gamma_2^{-1} (bu)^{2-\delta} \ge \gamma_1^{-1} u^{2-\delta}.$$

Because a copula function $C(u_1, u_2)$ is non-decreasing in both arguments, substituting these inequalities yields the following sandwich relationship for any $u > 0$:

$$C\big(\gamma_2 (au)^\delta, \gamma_2^{-1} (au)^{2-\delta}\big) \le C\big(\gamma_1 u^\delta, \gamma_1^{-1} u^{2-\delta}\big) \le C\big(\gamma_2 (bu)^\delta, \gamma_2^{-1} (bu)^{2-\delta}\big). \tag{1}$$

Step 2: Constructing the Logarithmic Inequality

Because the limit λ exists as a finite value, the copula expression $C(\gamma_2 v^\delta, \gamma_2^{-1} v^{2-\delta})$ must be strictly positive for all sufficiently small values of $v > 0$. Let $v_0 > 0$ be the threshold below which this positivity holds.

We restrict our analysis to $u \in (0, u_0)$ such that $bu < v_0$. Within this region, all three quantities in (1) are strictly positive. Since the natural logarithm is a monotonically increasing function on $(0, \infty)$, we can apply it across the inequality:

$$\log C\big(\gamma_2 (au)^\delta, \gamma_2^{-1} (au)^{2-\delta}\big) \le \log C\big(\gamma_1 u^\delta, \gamma_1^{-1} u^{2-\delta}\big) \le \log C\big(\gamma_2 (bu)^\delta, \gamma_2^{-1} (bu)^{2-\delta}\big).$$

For $u \in (0,1)$, $\log u$ is strictly negative. Dividing the entire inequality by $\log u$ reverses the direction of the signs:

$$\frac{\log C\big(\gamma_2(au)^\delta, \gamma_2^{-1}(au)^{2-\delta}\big)}{\log u} \geq \frac{\log C\big(\gamma_1 u^\delta, \gamma_1^{-1}u^{2-\delta}\big)}{\log u} \geq \frac{\log C\big(\gamma_2(bu)^\delta, \gamma_2^{-1}(bu)^{2-\delta}\big)}{\log u}. \tag{2}$$

Step 3: Evaluating the Limits

Let $w > 0$ represent an arbitrary positive constant (which will stand in for a and b). Consider the sequence:

$$B_w(u) = \frac{\log C\big(\gamma_2(wu)^\delta, \gamma_2^{-1}(wu)^{2-\delta}\big)}{\log u}.$$

We can algebraically rewrite $B_w(u)$ as the product of two terms:

$$B_w(u) = \frac{\log C\big(\gamma_2(wu)^\delta, \gamma_2^{-1}(wu)^{2-\delta}\big)}{\log(wu)} \cdot \frac{\log(wu)}{\log u}.$$

As $u \downarrow 0$, the scaled variable $(wu) \downarrow 0$. By the initial assumption defining λ, the first term evaluates to:

$$\lim_{u\downarrow 0} \frac{\log C\big(\gamma_2(wu)^\delta, \gamma_2^{-1}(wu)^{2-\delta}\big)}{\log(wu)} = \lambda.$$

For the second term, we apply logarithmic expansion:

$$\lim_{u\downarrow 0} \frac{\log(wu)}{\log u} = \lim_{u\downarrow 0} \left(1 + \frac{\log w}{\log u}\right).$$

Since w is a constant, $\log w$ is fixed, and as $\log u \to -\infty$, the fraction $\frac{\log w}{\log u}$ converges to 0. Thus, the second term evaluates to 1.

Therefore, for any fixed constant $w > 0$, we have:

$$\lim_{u\downarrow 0} B_w(u) = \lambda \cdot 1 = \lambda.$$

Applying this result to both our lower bound (where $w = b$) and our upper bound (where $w = a$) in (2), we see that the outer terms of our reversed inequality both converge to λ. By the Squeeze Theorem, the central term must also converge to λ:

$$\lim_{u\downarrow 0} \frac{\log C\big(\gamma_1 u^\delta, \gamma_1^{-1}u^{2-\delta}\big)}{\log u} = \lambda.$$

This shows that $\kappa_L(\delta, \gamma_1; C) = \kappa_L(\delta, \gamma_2; C)$. Because $\delta \in (0,2)$ and $\gamma_1, \gamma_2 > 0$ were chosen arbitrarily, the value $\kappa_L(\delta, \gamma; C)$ is independent of the scaling parameter γ. □

Verification for the Pareto-Mixture Copula (C_{PM})

By Definition 1, the function C_{PM} represents the underlying dependence structure of the continuous random variables (X_1, X_2). According to Sklar's Theorem, the function derived via

$$C_{\mathrm{PM}}(u_1, u_2) = F_{X_1,X_2}\big(F_{X_1}^{-1}(u_1), F_{X_2}^{-1}(u_2)\big)$$

is mathematically guaranteed to be a valid copula on $[0,1]^2$.

Because C_{PM} is a proper copula, it strictly obeys the property that it is non-decreasing in each coordinate, which was the foundational assumption required for the proof above. Consequently, provided that the limit $\kappa_L(\delta, \gamma; C_{\mathrm{PM}})$ exists for any single chosen $\gamma > 0$, the exact same mathematical proof applies. The theoretical measure of tail behavior for the Pareto-mixture class relies entirely on the direction parameter δ and remains perfectly invariant to the chosen scaling parameter γ.